

SAI DHIREN MUSALOJI

AI Systems Engineer | Full-Stack & Applied AI Engineer

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SUMMARY

AI Systems Engineer with production experience spanning agentic AI, LLM infrastructure, and full-stack delivery. Designs and ships complete systems end-to-end, backend/API design (FastAPI, Node.js/Express, REST, GraphQL), data layers (PostgreSQL, Redis), orchestration and tool integration, and front-end interfaces (React, Next.js/SSR, TypeScript), with a focus on closing the gap between AI research and reliable, cost-defensible production systems. Comfortable owning system design from scratch: cloud infrastructure (AWS, Azure, GCP), CI/containerized deployment (Docker, Kubernetes, Terraform), and AI-assisted development tooling (Claude Code, Cursor) to accelerate implementation, testing, and debugging.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript

Frontend: React, Next.js (SSR), TypeScript, JavaScript, HTML, CSS, component-based/responsive UI development

Backend & APIs: FastAPI, Node.js/Express, RESTful API development & design, GraphQL, service orchestration, MCP-style tool gateways, WebSockets, Streamlit

AI & LLM Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, DSPy, Model Context Protocol (MCP)

LLM Providers & APIs: OpenAI, Azure OpenAI, Anthropic Claude, Google Vertex AI, NVIDIA Nemotron, HuggingFace Transformers, Ollama, Mistral AI

Vector Databases & Search: Pinecone, ChromaDB, FAISS, Weaviate, Qdrant, PGVector, Azure AI Search

ML & Data Tools: PyTorch, TensorFlow, Scikit-learn, XGBoost, PySpark, Pandas, NumPy, MLflow, Weights & Biases

Cloud & Infrastructure: AWS (S3, EC2, SQS, Rekognition, IAM, VPC), Azure AI Services, Google Cloud Platform, Docker, Kubernetes, Terraform (IaC)

Data & Backend Systems: PostgreSQL, Redis, Apache Spark, Hadoop/MapReduce, Apache Oozie, Apache Kafka, Databricks

Architecture & Practices: Component-based / modular architecture, event-driven & asynchronous design, distributed systems fundamentals, agent-to-agent communication and tool orchestration, responsive/accessible UI development (React, Next.js, Streamlit)

AI-Assisted Development: Claude Code, Cursor (used to accelerate implementation, testing, and debugging workflows)

EXPERIENCE

AI Engineer | *Tech Mahindra Americas*

Mar 2026 – Present

- **Architected a multi-agent conversational platform (“Refill”)** on FastAPI and PostgreSQL: stage-aware billing/refill journeys, multi-agent orchestration, and a RESTful, MCP-style tool gateway (registered tools, invoke/audit path, conversation audit tables), mirroring service-oriented backend architecture with clear API contracts and traceable tool calls.
- **Implemented persisted state and context management** in PostgreSQL (turn history, dialog slots, TTL expiry, PII-aware sanitization, capped retention, aggregated usage metering) to keep multi-step, chained API calls bounded, auditable, and cost-efficient in production.
- **Built intent-aware request routing** (heuristic vs. LLM-first / hybrid modes) with planner integration and graceful fallbacks, a pattern directly analogous to backend request routing and API gateway design, optimizing for latency and cost per request.
- **Designed model-agnostic engineering and deployment adapters** (prompt normalization, vendor schema mapping, validation, auth/cost/SLA layers) to reduce vendor lock-in and support component swaps and A/B-style rollout across the stack.
- **Owned system design end-to-end**, building the multi-provider demo/admin UI as a React + Next.js (SSR) application in TypeScript, consuming the backend through a mix of REST and GraphQL endpoints for tool metadata, conversation history, and knowledge-context retrieval.
- **Operated local and private inference infrastructure** on external NVIDIA GPU hardware, with internal hosting for sensitive workloads, ensuring proprietary data and prompts stay off public APIs per policy, hands-on experience with deployment, environment provisioning, and infra-level troubleshooting.

AI/ML Engineer | *Tawin Solutions LLC (USA)*

Sept 2025 – Feb 2026

- **Designed a personalized recommendation engine** (collaborative + content-based filtering) incorporating user preference history, geolocation signals, and contextual features to improve suggestion relevance for an end-user-facing product.
- **Integrated Azure AI services with a structured logging/telemetry pipeline** to capture real-time user interactions and venue data, enabling continuous feedback loops for model retraining and drift detection, comparable to monitoring/telemetry practices used to track application health and usage in production.
- **Engineered feature pipelines from behavioral/event data**, translating raw interaction history into embeddings and structured signals consumed downstream by the recommendation service.

- **Contributed to Project Indus**, India's first multilingual LLM for native Indian languages; built data engineering pipelines (collection, cleaning, formatting of low-resource language corpora) and supported transformer architecture work during pre-training.
- **Built a real-time multilingual transcription service** for German-to-English meeting audio using Python-based ASR/NLP tooling, optimized for low-latency streaming processing and domain-specific vocabulary accuracy.

- **Provisioned and automated cloud infrastructure** across AWS (EC2, S3, IAM, VPC), Azure, and GCP using Terraform, writing modular Infrastructure-as-Code configurations for reproducible environment setup across deployment targets.

PROJECTS

Intelligent Research Copilot: Full-Stack RAG Platform

- Built an end-to-end research automation platform with a FastAPI backend exposing REST and GraphQL endpoints for ingestion, semantic retrieval, and abstractive summarization, paired with a React/Next.js (TypeScript) front-end, and a Streamlit variant for rapid internal iteration. Designed a modular, component-based pipeline architecture separating ingestion, retrieval, and presentation layers for maintainability and scale.

Real-Time Ride Matching Engine: Distributed Backend System

- Engineered a low-latency rider-driver matching backend using Redis geospatial indexing, PostgreSQL state management, and WebSocket-based real-time updates for live client communication. Prototyped distributed, event-driven coordination and agent-to-agent assignment flows, directly applying agentic orchestration and tool/service coordination patterns to a production-style mobility backend.

Event-Driven Vision Pipeline on AWS: Cloud-Native Backend Service

- Built a cloud-native detection pipeline integrating Amazon S3, Rekognition, and SQS behind an asynchronous service layer for large-batch, event-driven processing. Implemented service orchestration and confidence-based filtering, demonstrating cloud-platform fluency (AWS) and asynchronous backend design applicable to any cloud-hosted API service.

City-Scale Traffic Forecasting: Spatiotemporal ML Service

- Developed a spatiotemporal forecasting system (LSTM/ARIMA) for short-horizon traffic speed prediction on sensor data, with preprocessing, training, and evaluation pipelines built as reusable, modular components for intelligent transportation applications.

Containerized Spark ML Pipeline: Wine Quality Prediction

- Designed a Dockerized PySpark pipeline for training and inference with cloud-compatible storage integration, building reproducible, containerized batch workflows suitable for CI/CD-driven deployment.

Distributed Flight Analytics with Hadoop + Oozie

- Implemented large-scale flight performance analytics using Hadoop MapReduce, orchestrating multi-stage batch jobs with Apache Oozie to deliver delay/on-time trend analysis across high-volume datasets.

AI-Powered Audio Intelligence and Source Separation

- Built an audio intelligence application for music analysis and source separation with an interactive front-end for visual analytics, combining signal processing with deep-learning-based separation workflows.

Personal Portfolio & Project Showcase (musalojdhiren.vercel.app)

- Designed and built a personal site with React and Next.js (SSR, TypeScript), covering component architecture, routing, and deployment on Vercel, used as the front-end shell for showcasing full-stack and AI systems work.

EDUCATION

Master of Science in Data Science, New Jersey Institute of Technology

Bachelor of Technology in Electronics & Communications, Mahatma Gandhi Institute of Technology

CERTIFICATIONS

- Azure AI Engineer Associate (Microsoft AI-102)
- Azure Data Scientist Associate (Microsoft DP-100)
- Databricks Generative AI Fundamentals (Databricks)